

Kirk MacKinnon

Post-doctoral Associate

May 2026

 Center for Genomics and Systems Biology, NYU

 kirkmackinnon.com

 km5767@nyu.edu

 kirkmackinnon

About me

Molecular and computational biologist studying gene regulatory networks using integrated genomic approaches (RNA-seq, DAP-seq, and related datasets). My work focuses on understanding transcriptional regulation while developing practical, reproducible approaches to analyzing complex biological data.

I am particularly interested in training students in applied bioinformatics, emphasizing clear biological interpretation alongside computational skills, and contributing to the development of collaborative, student-centered research and teaching programs.

My training spans end-to-end experimental and computational genomics, from DNA/RNA extraction and sequencing library preparation to high-throughput data analysis and modeling.

Education

- 2015–2021 Doctorate of Philosophy
Molecular & Cellular Biology UMass Amherst
- Transcriptional regulation and gene regulatory networks; emphasis on genomic data analysis
- 2011–2015 Bachelor of Science
Biology UMass Amherst

Research Experience

- 2022–Present Postdoctoral Associate Coruzzi Lab
New York, NY
- Develop and analyze cell-specific gene regulatory networks using RNA-seq and DAP-seq datasets
 - Integrate computational and experimental approaches to identify regulatory interactions and network structure
 - Design reproducible analysis pipelines for high-throughput genomic data
 - Mentor students in computational and biological data analysis
- 2015–2021 Ph.D. Researcher Hazen Lab
Amherst, MA
- Investigated transcriptional regulation of growth in *Brachypodium distachyon*
 - Analyzed RNA-seq and time-series datasets to identify regulatory patterns and environmental responses
 - Contributed to multiple peer-reviewed publications
 - Developed experience in experimental design and data interpretation
- 2013–2015 Research Assistant Hazen Lab
Amherst, MA
- Performed time-lapse image analysis of plant growth under varying abiotic conditions
 - Assisted in data collection and analysis for ongoing lab projects

Teaching

- 2020 Bioinformatics Workshop Instructor (Lab-based) Hazen Lab
Amherst, MA
- Designed and delivered an introductory bioinformatics course for graduate researchers
 - Taught foundational skills including Bash, R, and general computational workflows
 - Developed structured lessons and assignments to support applied learning
 - Mentored participants in applying computational methods to their research projects
- 2018–2020 Graduate Teaching Assistant – Introductory Biology I (BIO 151)
University of Massachusetts Amherst
Amherst, MA
- Led discussion sections and supported student learning in foundational biology concepts
 - Assisted in grading and evaluation of student work
 - Facilitated understanding of experimental design and data interpretation
 - Mentored undergraduate students in academic and scientific development

Publications

- In prep. Landscape of Transcription Factor Regulatory Harmony in Arabidopsis
Kirk J.-M. MacKinnon et al.
- Submitted Genome-wide transcription factor binding reveals regulatory logic and evolutionary divergence in grasses *Nature Communications*
Bahman Khahani, Leo A. Baumgart, Kirk R. Amundson, Ian W. McCahill, Greg A. Gregory, Kirk J.-M. MacKinnon, Xiaocheng Yu, Ritesh Mewalal, Anthony Bortolazzo, Yu Zhang, Madelaine E. Bartlett, John P. Vogel, Ian K. Blaby, Ronan C. O'Malley, Samuel P. Hazen
- Submitted Combinatorial effects of transcription factors functionally revealed by single cell TARGET-seq *Molecular Systems Biology*
Amélie Emanuel, Jean-Baptiste Carluer, Will E. Hinckley, Baldwin Dumortier, Samantha Frangos, Kirk MacKinnon, Bruno Guillotin, Tim L. Jeffers, Yitao Long, Manpreet Katari, Dennis Shasha, Angelo Pasquino, Matthew D. Brooks, Stéphanie Pateyron, Myriam Boyer-Clavel, Stéphanie Boireau, Wojciech Szponarski, Hatem Rouached, Sandrine Ruffel, André Mas, Gloria Coruzzi, Bastiaan O. Bargmann, Etienne Delannoy
- 2024 Mechanically induced localisation of SECONDARY WALL INTERACTING bZIP is associated with thigmomorphogenic and secondary cell wall gene expression *Quantitative Plant Biology*
Joshua H. Coomey, Kirk J.-M. MacKinnon, Ian W. McCahill, Bahman Khahani, Pubudu P. Handakumbura, Gina M. Trabucco, Jessica Mazzola, Nicole A. Leblanc, Rithany Kheam, Miriam Hernandez-Romero, Kerrie Barry, Lifeng Liu, Ji E. Lee, John P. Vogel, Ronan C. O'Malley, James J. Chambers, Samuel P. Hazen
• <https://doi.org/10.1017/qpb.2024.5>
- 2021 Mechanically induced localization of SECONDARY WALL INTERACTING bZIP is associated with thigmomorphogenic and secondary cell wall gene expression
Joshua H. Coomey, Kirk J.-M. MacKinnon, Ian W. McCahill, Bahman Khahani, Pubudu P. Handakumbura, Gina M. Trabucco, Jessica Mazzola, Nicole A. Leblanc, Rithany Kheam, Miriam Hernandez-Romero, Kerrie Barry, Lifeng Liu, Ji E. Lee, John P. Vogel, Ronan C. O'Malley, James J. Chambers, Samuel P. Hazen
• <https://doi.org/10.1101/2021.02.03.429573>
- 2020 Changes in ambient temperature are the prevailing cue in determining *Brachypodium distachyon* diurnal gene regulation *New Phytologist*
Kirk J.-M. MacKinnon, Benjamin J. Cole, Chang Yu, Joshua H. Coomey, Nolan T. Hartwick, Marie-Stanislas Remigereau, Tomás Duffy, Todd P. Michael, Steve A. Kay, Samuel P. Hazen
• <https://doi.org/10.1111/nph.16507>
- 2018 SECONDARY WALL ASSOCIATED MYB1 is a positive regulator of secondary cell wall thickening in *Brachypodium distachyon* and is not found in the Brassicaceae *The Plant Journal*
Pubudu P. Handakumbura, Kathryn Brow, Ian P. Whitney, Kangmei Zhao, Karen A. Sanguinet, Scott J. Lee, Jennifer Olins, Sandra P. Romero-Gamboa, Michael J. Harrington, Carlisle J. Bascom, Kirk J.-M. MacKinnon, Michael T. Veling, Lifeng Liu, Ji E. Lee, John P. Vogel, Ronan C. O'Malley, Magdalena Bezanilla, Laura E. Bartley, Samuel P. Hazen
• <https://doi.org/10.1111/tpj.14047>
- 2018 Secondary Wall Regulating NACs Differentially Bind at the Promoter at a CELLULOSE SYNTHASE A4 Cis-eQTL *Frontiers in Plant Science*
Jennifer R. Olins, Li Lin, Scott J. Lee, Gina M. Trabucco, Kirk J.-M. MacKinnon, Samuel P. Hazen
• <https://doi.org/10.3389/fpls.2018.01895>
- 2014 Daily Changes in Temperature, Not the Circadian Clock, Regulate Growth Rate in *Brachypodium distachyon* *PLoS ONE*
Dominick A. Matos, Benjamin J. Cole, Ian P. Whitney, Kirk J.-M. MacKinnon, Steve A. Kay, Samuel P. Hazen
• <https://doi.org/10.1371/journal.pone.0100072>

Awards

2021	Howard E. and Margaret E. Barr Bigelow Memorial Scholarship.
2021	Lotta M. Crabtree Fellowship in Production Agriculture.
2020	Center for Agriculture, Food and the Environment Fellowship.
2019	Lotta M. Crabtree Fellowship in Production Agriculture.
2018	Center for Agriculture, Food and the Environment Fellowship.
2015	Outstanding Biology Research Award.

Conference Presentations

Talks

2018	Exploring Diurnal Rhythmic Gene Expression in <i>Brachypodium distachyon</i> . ASPB Northeastern Section Meeting.
------	---

Posters

2019	Changes in ambient temperature are the prevailing cue in determining <i>Brachypodium distachyon</i> diurnal gene regulation. Annual Symposium in Plant Biology.
2019	Exploring Diurnal Rhythmic Gene Expression in <i>Brachypodium distachyon</i> . PAG XXVII.
2014	Daily Changes in Temperature, Not the Circadian Clock, Regulate Growth Rate in <i>Brachypodium distachyon</i> . UMass Undergraduate Research Symposium.

Mentoring & Outreach

Train and mentor undergraduate assistants in the Hazen Lab

2017–2021	Undergraduate Research Mentorship University of Massachusetts Amherst	Hazen Lab
• Mentored undergraduate students in plant biology research projects • Provided guidance in experimental design and data analysis • Supported development of scientific reasoning and communication skills • Students mentored: Sanyam Shilpakar; Chang Yu; Alden Nichols; Jacob Scott; Crysta Daniels		

Creation of online tools for data exploration

2018–Present	Development of Online Genomic Data Exploration Tools	Hazen Lab; Coruzzi Lab
• Developed interactive Shiny applications for exploration of RNA-seq datasets • Designed tools to make complex genomic data accessible to researchers • Built an online platform for exploring transcription factor regulatory harmony data • Applications available at: hazenlab.shinyapps.io		